Day - 11
Python - For Loop

SKILL COURSE

A.I. Powered 30 Days Python Micro Course By Satish Dhawale (Microsoft Certified Trainer)

What is a Loop?

- A loop means **doing the same work again and again automatically**.

- **Example:**
  - f you want to say “Hello” 10 times, you don't type print("Hello") 10 times. You use a loop.

- A loop repeats a task for you — **no manual repetition**.

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Technical Definition

- A **for loop** is used to iterate (repeat) over a sequence such as:
  - list
  - string
  - range
  - tuple
  - dictionary
- Python executes the loop block **for each item** in the sequence.

- Syntax of For Loop

```
for variable in sequence:
    # repeated code
```

Why For Loop is Important for Data Analysts?

Process	• Process each row in a dataset
Clean	• Clean data in loops
Apply	• Apply operations on multiple values
Split	• Split large files
Summarize	• Summarize values
Replace	• Replace multiple wrong spellings
Read	• Read CSV lines
Automate	• Automate repetitive tasks

Example

Print 1 to 5

- for i in range(1, 6):
- print(i)

Print “Hello” 5 Times

- for i in range(5):
- print("Hello")

Looping Over a List

- fruits = ["Apple", "Banana", "Mango"]
- for item in fruits:
- print(item)

Exmaple

Looping Over a String (Character by Character)

- word = "Python"
- for letter in word:
- print(letter)

Loop with Range (Start, End, Step)

- for n in range(0, 20, 2):
- print(n)

Practical Example – Total Marks Calculation

```
marks = [78, 82, 90, 69]

total = 0
for m in marks:
    total += m

print("Total Marks:", total)
```

Real Data Analyst Example – Cleaning City Names

```
cities = ["mUMbai", "  DELhi  ", "pune", "Chennai "]

cleaned = []

for c in cities:
    cleaned.append(c.strip().title())

print(cleaned)
```

Data Analyst Example – Fixing Spelling Mistakes

```
wrong_list = ["Bengluru", "Mombai", "Kolkatta"]
correct_list = []

for city in wrong_list:
    correct_list.append(
        city.replace("Bengluru", "Bengaluru")
            .replace("Mombai", "Mumbai")
            .replace("Kolkatta", "Kolkata")
    )

print(correct_list)
```

For Loop with If Condition

```
numbers = [5, 12, 3, 18, 7]

for n in numbers:
    if n > 10:
        print(n, "is greater than 10")
```

Looping Through Dictionary

```
student = {"name": "Satish", "age": 25, "city": "Pune"}

for key, value in student.items():
    print(key, ":", value)
```

Advanced Example – Extract Last 4 Digits of Each ID

```
ids = ["EMP-001122", "EMP-889900", "EMP-550012"]

for emp in ids:
    print(emp[-4:])
```

Assignments for Students

✓ Basic	✓ Intermediate	✓ Advanced
Print numbers from 1 to 20 Print each letter of your name	Clean city names list Count vowels in a word	Extract last 3 digits from multiple IDs Calculate total, average from marks list